

1. Overview of Physical Security

- Physical security involves protecting physical assets such as buildings, hardware, personnel, and infrastructure from physical threats.
 - It is the first layer of defense in an organization's security framework.
 - Objectives include preventing unauthorized access, theft, damage, or disruption.
 - It covers barriers like fences, locks, security guards, surveillance cameras, lighting, and environmental controls.
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2. Need for Physical Security

- Physical security is critical to protect IT infrastructure and sensitive information.
 - Prevents theft, sabotage, vandalism, espionage, and natural disasters.
 - Physical breaches often lead to data breaches or operational downtime.
 - Regulatory compliance (e.g., GDPR, HIPAA) often mandates physical controls.
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3. Factors Affecting Physical Security

- **Location:** Urban vs. remote areas affect risks like burglary or natural disasters.

- **Environment:** Climate, natural disasters (earthquakes, floods).
 - **Building Design:** Entry points, surveillance coverage, secure zones.
 - **Personnel:** Training and awareness of staff.
 - **Technology:** Use of access control systems, alarms, and monitoring.
 - **Policy:** Security policies and procedures.
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4. Penetration Testing Methodologies

Penetration testing (pen testing) simulates cyberattacks to identify vulnerabilities.

Common Methodologies:

- **Reconnaissance:** Gathering information (passive & active).
- **Scanning:** Using tools to identify open ports, services, and vulnerabilities.
- **Gaining Access:** Exploiting vulnerabilities to enter systems.
- **Maintaining Access:** Establishing backdoors or persistence.
- **Analysis & Reporting:** Documenting findings with remediation recommendations.

Popular frameworks include:

- OSSTMM: Open Source Security Testing Methodology Manual.
- OWASP PTES: Penetration Testing Execution Standard.
- NIST SP 800-115: Technical guide for information security testing.